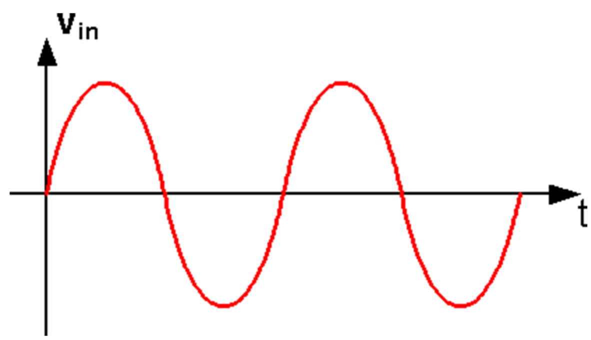
SC2017/CE2107 Microprocessor System Design and Development

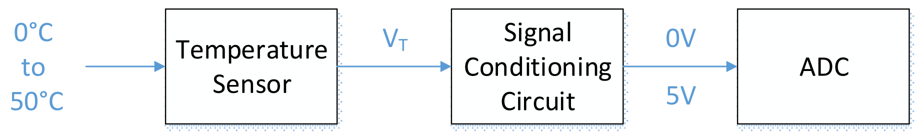
Tutorial 5

<u>Sensor and Signal Conditioning, Analog Interface</u>

1. Plot the voltage vs. time of a non-inverting comparator and an inverting comparator for the following input voltage. The reference voltage for both comparators is 0V.

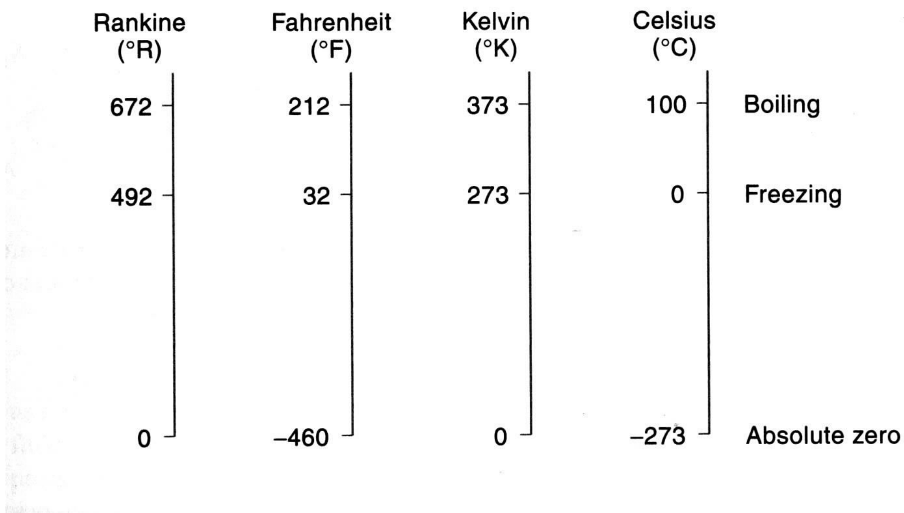


2. You are required to design a signal conditioning circuit to interface between a temperature sensor and the ADC. The temperature range to be measured is 0°C to 50°C. The range of the ADC converter is 0 to 5V.



The output of the signal conditioning circuit should be linear; i.e. when the sensor is measuring 0°C, the output of the signal conditioning circuit should be 0V; when the sensor is measuring 25°C, the output of the signal conditioning circuit should be 2.5V; when the sensor is measuring 50°C, the output of the signal conditioning circuit should be 5V.

The temperature sensor has a sensitivity of 10mV/°K. The relationship between degrees Kelvin and Celsius is: 1-degree rise in Kelvin equals a 1-degree rise in Celsius. The freezing point of water is 0°C, which equals to 273°K.



a. Write an equation that describes the input-output characteristic of the temperature sensor (in terms of V_T and the measured temperature in Celsius, T_C).

Vt = (Tc + 273) * 10 * 10^-3

b. Find the output voltage range of the temperature sensor (V_T).

0 to 50 C == 2.73 to 3.23 V

c. Find the gain that is required to amplify V_T to fit the input range of the ADC. Using inverting amplifiers, draw the block diagram of the signal conditioning circuit to realize this gain.

(5-0)/(3.23-2.73) = 10 gain. Usually use negative(inverting) amplifier circuit, if want positive again, use another inverting amplifier.

d. Explain why the circuit is not complete. Write a linear equation for the signal conditioning circuit output voltage to satisfy the requirements of the application.

Vout = 10xVt + Voffset, the reason being 10Vt = 27.3 to 32.3V which is too hight, need to offset it to be 0 - 5V

e. Briefly explain how you can modify the solution from 1(c) to implement the final circuit. Draw a block diagram of the final circuit.

inverting sum amplifier, to add multiple input signals together and produce a weighted sum output, the gain of each signal can be individually adjuseted by choosing a different resistor value for each input. Gain1 = -10, Gain2 = -1.82.

f. If a 3-bit ADC is used and the output generated by the ADC is 101_2, determine the measured temperature in Celsius T_C.

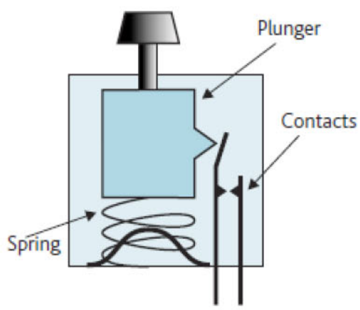
size of each bit, (5-0)/8 = 0.625V, 5x0.625 = 3.125V, Vt = (3.125+27.3) = 3.0425V, Tc = 3.0425/(10*10^-3) - 273 = 31.25 C

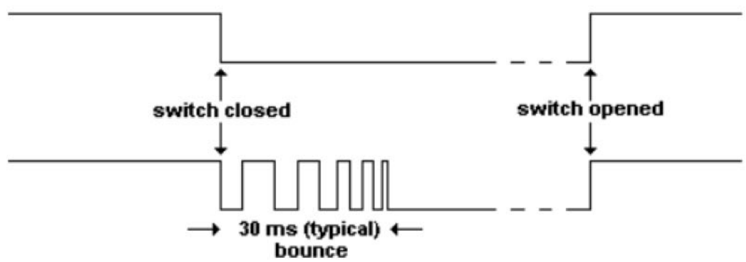
3. Answer the following questions on the hardware successive approximation ADC and digital ramp ADC. 3a SAR is generally faster
   a. State the benefit of the successive approximation hardware converter over digital ramp ADC.
   b. Explain how this benefit is achieved by describing the working principle of the successive approximation converter. SAR changes MSB than slowly goes down the bits to LSB
   c. For an m-bit analogue-to-digital conversion, state the maximum number of iterations required for both the successive approximation and digital ramp ADC.

SAR maximum is m iterations, Digital ramp has max iterations of 2^m

Optional

4. Switches suffer from Switch Bounce, which produces a series of pulses when contact is made. This can be registered as a series of ON and OFF signals lasting several milliseconds instead of just the one intended single and positive switching action. The easiest software approach for debouncing is Wait and See (see pseudo code below).






```
Set Return_Value = !Switch_Pressed

If (Switch_Pin == 0) // Switch is pressed
{
      WAIT (DEBOUNCE_PERIOD);
      If (Switch_Pin == 0) // If switch is still pressed
      {
            while (Switch_pin == 0);
            Return_Value = Switch_Pressed;
      }
}
return (Return_Value);
```

a) In certain applications, a single keypress is registered only upon the release of the button. How would you modify the pseudo-code of Wait and See above, to register the key press only when the button has been released?

b) Discuss a practical problem of your solution in a.